

Starting data (after mapping and hit count analysis)

	t1	t2	t3	c1	c2	c3	pseudo-sample	
gene1	100/0.94	120/81.83	110/81.83	50/81.83	65/81.83	70/81.83	geom_mean(100,120, 110, 50,65,70) =	
500bp	81.83							
gene2	250/0.94	260/265.66	270/265.66	265/265.66	270/265.66	280/265.66	265.66	
3kb								
gene20000								

median(1.22,0.94,..)
=0.94 → normalization factor for sample t1
Total reads 1Mio 1.5M 1M 300k 450k 500k

Final step: divide each raw count value by the normalization factor

$$TPM_i = 10^6 \times (n_i / L_i) / \sum (n_j / L_j)$$
$$TPM_{gene1(t1)} = 10^6 \times 0.2 / (0.2 + 250 / 3000 \dots)$$

assuming we have only gene1 and gene2 as genes in the organism of our interest

$$TPM_{gene1(t1)} = 1M \times 0.2 / 0.283 = 706713$$
$$TPM_{gene2(t1)} = 1M \times 0.083 / 0.283 = 293287$$

$$FPKM = (no\ of\ reads \times 10^6) / (total\ no\ of\ reads \times length_gene)$$
$$FPKM_{gene1(t1)} = 100 \times 10^6 / (10^6 \times 500) = 0.2$$
$$FPKM_{gene2(t1)} = 250 \times 10^6 / (10^6 \times 3000) = 0.083$$